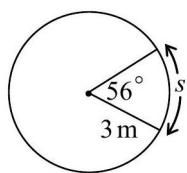


mb-pc40s-2015-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Use the information in the diagram to determine the value of the arc length " s ", given the central angle of 56° .



Solution

$$\theta = 56^\circ \times \frac{\pi}{180^\circ}$$

$$\theta = \frac{56\pi}{180} \text{ or } \frac{14\pi}{45}$$

$$s = \theta r$$

$$s = \left(\frac{14\pi}{45} \right) (3\text{ m})$$

$$s = \frac{14\pi}{15} \text{ m or } 2.932 \text{ m}$$

1 mark for conversion

1 mark for substitution

2 marks

Question 2

Solve $\tan^2 \theta - 5 \tan \theta + 4 = 0$ where $\theta \in \mathbb{R}$.

Solution

Method 1

$$(\tan \theta - 1)(\tan \theta - 4) = 0$$

$$\tan \theta = 1 \quad \tan \theta = 4$$

$$\theta_r = 1.3258$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4} \quad \theta = 1.326, 4.467$$

$$\theta = 0.785, 3.927$$

1 mark for solving for $\tan \theta$ ($\frac{1}{2}$ mark for each branch)

2 marks ($\frac{1}{2}$ mark for each value of θ)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + 2k\pi \\ &= 1.326 + 2k\pi \\ &= \frac{5\pi}{4} + 2k\pi \\ &= 4.467 + 2k\pi \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

1 mark for general solution

4 marks

Method 2

$$(\tan \theta - 1)(\tan \theta - 4) = 0$$

$$\tan \theta = 1 \quad \tan \theta$$

$$\theta_r = 1.3258$$

$$\theta = \frac{\pi}{4} \theta = 1.326$$

$$\theta = 0.785$$

1 mark for solving for $\tan \theta$ ($\frac{1}{2}$ mark for each branch)

2 marks (1 mark for each value of θ)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + k\pi \\ &= 1.326 + k\pi \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

1 mark for general solution

4 marks

Question 3

Solve:

$$2^{5x} = 3(5)^{x-3}$$

Solution

$$5x \log 2 = \log 3 + (x - 3) \log 5$$

$$5x \log 2 = \log 3 + x \log 5 - 3 \log 5$$

$$5x \log 2 - x \log 5 = \log 3 - 3 \log 5$$

$$x(5 \log 2 - \log 5) = \log 3 - 3 \log 5$$

$$x = \frac{\log 3 - 3 \log 5}{5 \log 2 - \log 5}$$

$$x = -2.009$$

$\frac{1}{2}$ mark for applying logarithms

1 mark for product rule

1 mark for power rule

$\frac{1}{2}$ mark for collecting like terms

$\frac{1}{2}$ mark for isolating x

$\frac{1}{2}$ mark for evaluating a quotient of logarithms

4 marks

Question 4

David and Sarah are in a class of 10 boys and 8 girls.

A committee of 3 boys and 2 girls is to be selected from the students in this class.

Determine the number of possible committees if David and Sarah cannot be on the same committee.

Solution

Method 1

All:	${}_{10}C_3 \times {}_8C_2 = 3360$	1 mark for all possible committees
Both:	${}_9C_2 \times {}_7C_1 = 252$	1 mark for both on the committee
	$3360 - 252 = 3108$	1 mark for subtraction of cases
		3 marks

Method 2

Case 1: David, not Sarah	${}_9C_2 \times {}_7C_2 = 756$	$\frac{1}{2}$ mark for Case 1
Case 2: Sarah, not David	${}_9C_3 \times {}_7C_1 = 588$	$\frac{1}{2}$ mark for Case 2
Case 3: Neither David nor Sarah	$756 + 588 + 1764 = 3108$	1 mark for Case 3
		1 mark for addition of cases
		3 marks

Question 5

In the binomial expansion of $\left(\frac{3}{x^2} - x^5\right)^{10}$, simplify the 7 th term.

Solution

$$t_7 = {}_{10}C_6 \left(\frac{3}{x^2}\right)^4 (-x^5)^6$$

$$t_7 = 210 \left(\frac{81}{x^8}\right) (x^{30})$$

$$t_7 = 17010x^{22}$$

2 marks (1 mark for ${}_{10}C_6$, $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponent)

3 marks

Question 6

A lake affected by acid rain has a pH of 4.4.

A person suffering from heartburn has a stomach acid pH of 1.2.

The pH of a solution is defined as $\text{pH} = -\log[\text{H}^+]$ where $[\text{H}^+]$ is the hydrogen ion concentration.

How many times greater is the hydrogen ion concentration of the stomach than that of the lake?

Express your answer as a whole number.

Solution

Method 1

$$\begin{aligned}\text{pH} &= -\log[\text{H}^+] \\ -\text{pH} &= \log[\text{H}^+] \\ [\text{H}^+] &= 10^{-\text{pH}} && 1 \text{ mark for exponential form} \\ \frac{[\text{H}^+]_{\text{stomach}}}{[\text{H}^+]_{\text{lake}}} &= \frac{10^{-1.2}}{10^{-4.4}} && 1 \text{ mark for comparison} \\ &= 10^{3.2} \\ &= 1584.9 \\ &= 1585 && \text{marks}\end{aligned}$$

Method 2

Lake

$$\begin{aligned}4.4 &= -\log[\text{H}^+] \\ -4.4 &= \log[\text{H}^+] \\ 10^{-4.4} &= [\text{H}^+]\end{aligned}$$

Stomach

$$\begin{aligned}1.2 &= -\log[\text{H}^+] \\ -1.2 &= \log[\text{H}^+] \\ 10^{-1.2} &= [\text{H}^+]\end{aligned}$$

$\frac{1}{2}$ mark for exponential form

$$\begin{aligned}\frac{[\text{H}^+]_{\text{stomach}}}{[\text{H}^+]_{\text{lake}}} &= \frac{10^{-1.2}}{10^{-4.4}} \\ &= 10^{3.2} \\ &= 1585\end{aligned}$$

1 mark for comparison

2 marks

Question 7

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$\cos 2\theta - 3 \sin \theta - 2 = 0$$

Solution

$$(1 - 2 \sin^2 \theta) - 3 \sin \theta - 2 = 0$$

$$-2 \sin^2 \theta - 3 \sin \theta - 1 = 0$$

$$2 \sin^2 \theta + 3 \sin \theta + 1 = 0$$

$$(2 \sin \theta + 1)(\sin \theta + 1) = 0$$

$$\sin \theta = -\frac{1}{2} \quad \sin \theta = -1$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6} \quad \theta = \frac{3\pi}{2}$$

1 mark for correct substitution of an appropriate identity

1 mark for solving for $\sin \theta$ ($\frac{1}{2}$ mark for each branch)

2 marks for solutions (1 mark for each branch; $\frac{1}{2}$ mark for each value in the left branch)

4 marks

Question 8

Explain how the value of n affects the behaviour of the graph of the polynomial function

$p(x) = (x + 3)(x - 1)^n$, as $p(x)$ approaches the x -intercept at $x = 1$.

Solution

If n is even, the graph will turn at the x -axis at $x = 1$.

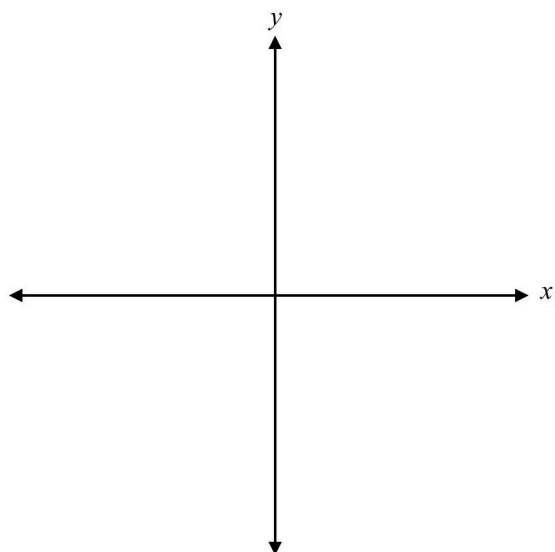
If n is odd, the graph will cross the x -axis at $x = 1$.

or

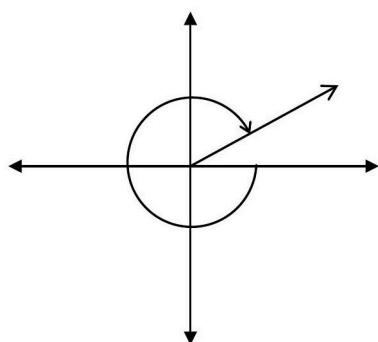
As n increases, $p(x)$ will become flatter around the intercept at $x = 1$.

Question 9

Sketch the angle -320° in standard position.



Solution



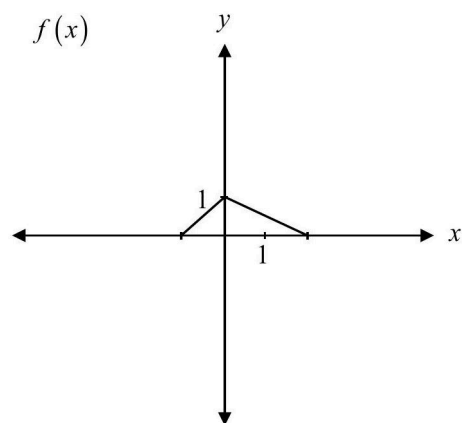
1 mark for angle drawn in quadrant I

1 mark

- If directional arrow is not indicated, it is a communication error, E1 (final answer not stated).

Question 10

Given the graph of $y = f(x)$, explain how to graph $y = f(-x)$.



Solution

Multiply the x -coordinates by -1 .

or

Reflect the graph over the y -axis.

Question 11

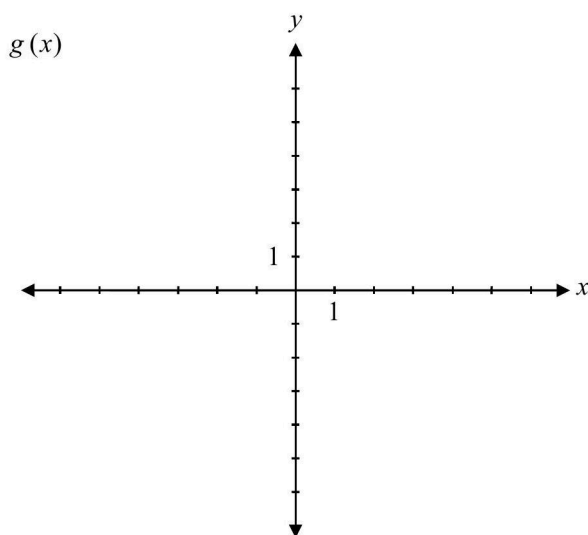
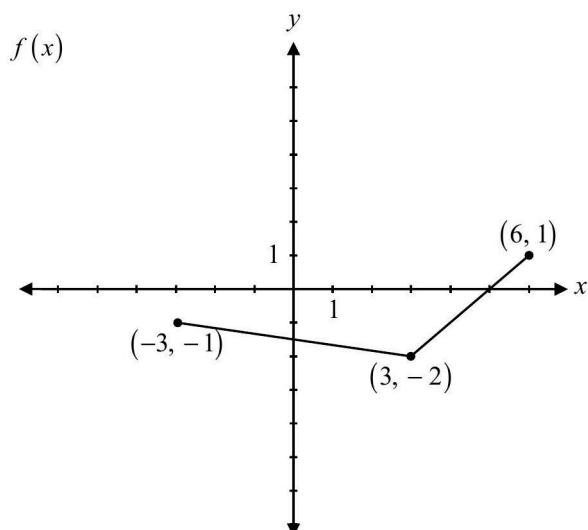
Explain how the graph of $y = \frac{3(x-1)}{(x-1)}$ is different than the graph of $y = 3$.

Solution

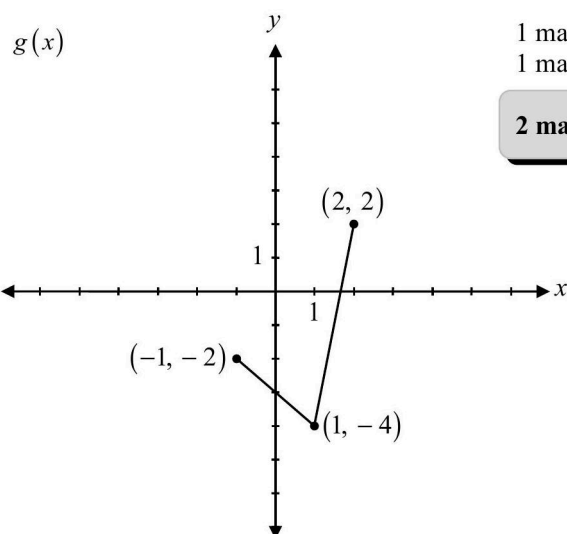
There is a point of discontinuity (hole) when $x = 1$ on the graph of $y = \frac{3(x-1)}{(x-1)}$. **1** mark

Question 12

Given the graph of $f(x)$, sketch the graph of $g(x) = 2f(3x)$.



Solution

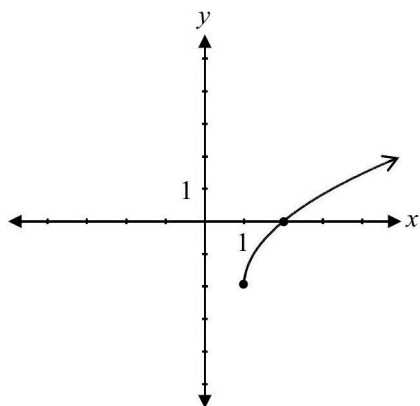


1 mark for horizontal stretch/compression
1 mark for vertical stretch

2 marks

Question 13

Determine the equation of the radical function represented by the graph.



Solution

$$y = 2\sqrt{x-1} - 2$$

or

1 mark for vertical stretch

1 mark for horizontal translation

1 mark for vertical translation

3 marks

$$y = \sqrt{4(x-1)} - 2$$

1 mark for horizontal compression

1 mark for horizontal translation

1 mark for vertical translation

3 marks

Question 14

There are 2 types of pencils, 3 colours of highlighters, and 5 styles of pens.

If you must select one of each to form a set, how many different sets of writing instruments are possible?

A.

10

B.

11

C.

25

D.

30

Solution

Correct answer: D

30

Question 15

The point $P(\theta)$ lies on the unit circle. What are the coordinates of the point P if $\theta = 120^\circ$?

A.

$$\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

B.

$$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$$

C.

$$\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

D.

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

Solution

Correct answer: A

$$\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

Question 16

Identify the function that has a domain of $\{x \mid x \geq 7\}$ and a range of $\{y \mid y \geq 0\}$.

A.

$$f(x) = \sqrt{x} + 7$$

B.

$$f(x) = \sqrt{x} - 7$$

C.

$$f(x) = \sqrt{x+7}$$

D.

$$f(x) = \sqrt{x-7}$$

Solution

Correct answer: D

$$f(x) = \sqrt{x-7}$$

Question 17

Using $y = -10 \cos[B(x - C)] + D$, the value of C that corresponds to the following graph is:

A.

5

B.

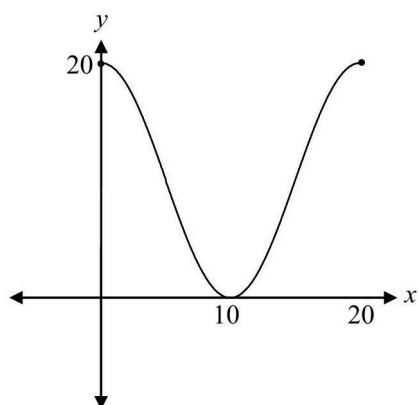
10

C.

15

D.

20



Solution

Correct answer: B

10

Question 18

Given the following row of Pascal's Triangle, identify the binomial expansion with these coefficients.

1 5 10 10 5 1

A.

$$(x + y)^4$$

B.

$$(x + y)^5$$

C.

$$(x + y)^6$$

D.

$$(x + y)^7$$

Solution

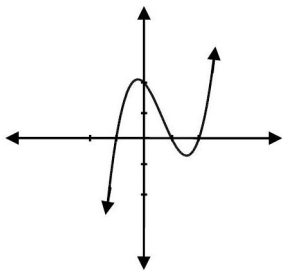
Correct answer: B

$$(x + y)^5$$

Question 19

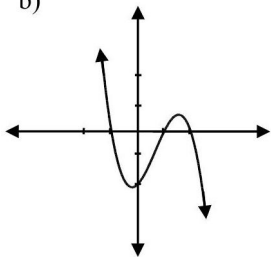
Identify the graph of the function $f(x) = -(x-2)(x-1)^2(x+1)$.

A.

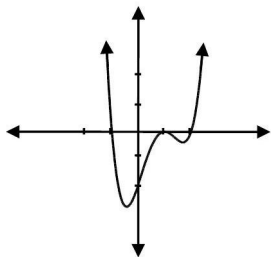


B.

B)

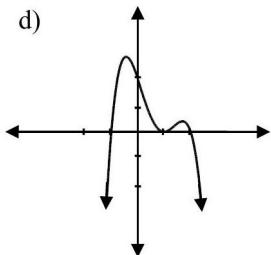


C.



D.

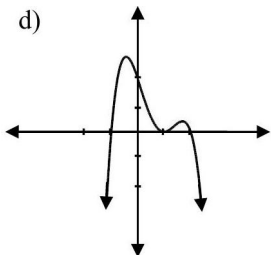
d)



Solution

Correct answer: D

d)



Question 20

Determine the value of $\log_9(\log_3 27)$.

A.

$$\frac{1}{3}$$

B.

$$\frac{1}{2}$$

C.

$$2$$

D.

$$3$$

Solution

Correct answer: B

$$\frac{1}{2}$$

Question 21

If (x, y) is a point on the graph of $y = f(x)$, identify the coordinates of this point on the graph of $g(x) = f(2x) + 5$.

A.

$$\left(\frac{x}{2}, y + 5\right)$$

B.

$$(2x, y + 5)$$

C.

$$\left(\frac{x}{2}, y - 5\right)$$

D.

$$\left(\frac{x}{2} - 5, y\right)$$

Solution

Correct answer: A

$$\left(\frac{x}{2}, y + 5\right)$$

Question 22

Identify an equivalent expression for $1 + \log_2 5$.

A.

$$\log_2 5$$

B.

$$\log_2 7$$

C.

$$\log_2 10$$

D.

$$\log_2 11$$

Solution

Correct answer: C

$$\log_2 10$$

Question 23

Solve:

$$2\log_4 x - \log_4(x+3) = 1$$

Solution

$$2\log_4 x - \log_4(x+3) = 1$$

$$\log_4\left(\frac{x^2}{x+3}\right) = 1$$

$$4^1 = \frac{x^2}{x+3}$$

$$4x + 12 = x^2$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \quad x > -2$$

1 mark for power rule

1 mark for quotient rule

1 mark for exponential form

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for rejecting extraneous root

4 marks

Question 24

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the y -axis
- horizontal translation of 3 units to the right
- vertical translation of 4 units down

Write the equation of $g(x)$ in terms of $f(x)$.

$$g(x) =$$

Solution

$$g(x) = f(-(x - 3)) - 4$$

or

1 mark for horizontal shift

1 mark for vertical shift

1 mark for reflection

3 marks

$$g(x) = f(-x + 3) - 4$$

Question 25

a) 3 marks

b) 1 mark

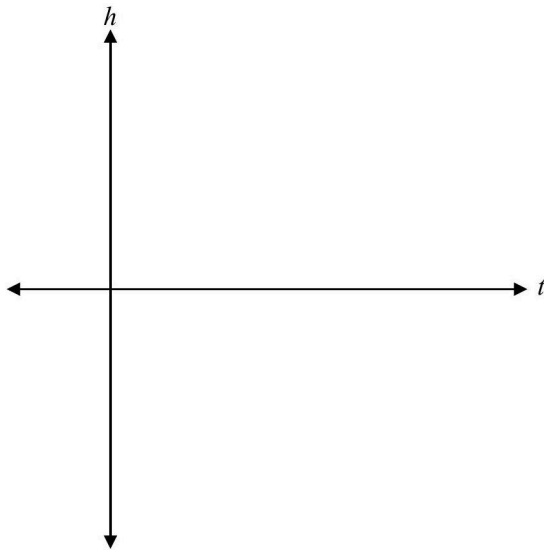
116 117

The height of a bicycle pedal as the bicycle is moving at a constant speed can be represented by the following function:

$$h(t) = 15 \cos \frac{2\pi}{5}t + 30$$

where h is the height of the pedal above the ground, in cm, and t is the time, in seconds.

a) Sketch a graph of at least one period of this function, where $t \geq 0$.

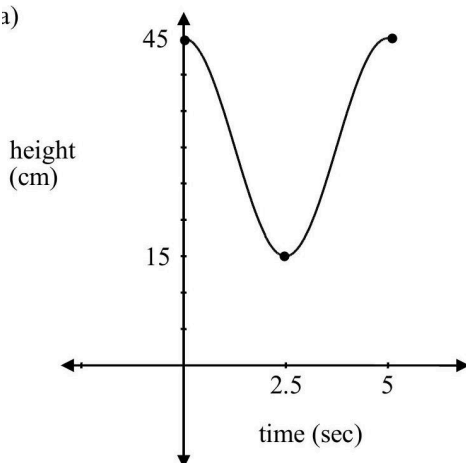


b) Determine the height of the bicycle pedal at 7.5 seconds.

Solution

• []

a)



Amplitude = 15 1 mark for amplitude

$$\begin{aligned} \text{Period} &= \frac{2\pi}{\frac{2\pi}{5}} \\ &= 5 \end{aligned}$$

1 mark for period

Vertical Shift = 30 1 mark for vertical shift

3 marks

b) From the graph:

$$h(t) = 15 \text{ cm}$$

or

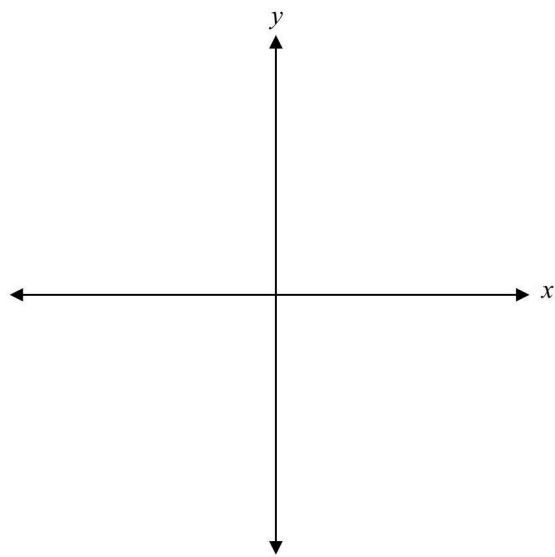
From the equation:

$$\begin{aligned}h(t) &= 15 \cos \frac{2\pi}{5}(7.5) + 30 \\&= 15 \cos 3\pi + 30 \\&= 15(-1) + 30 \\&= 15 \text{ cm}\end{aligned}$$

Question 26

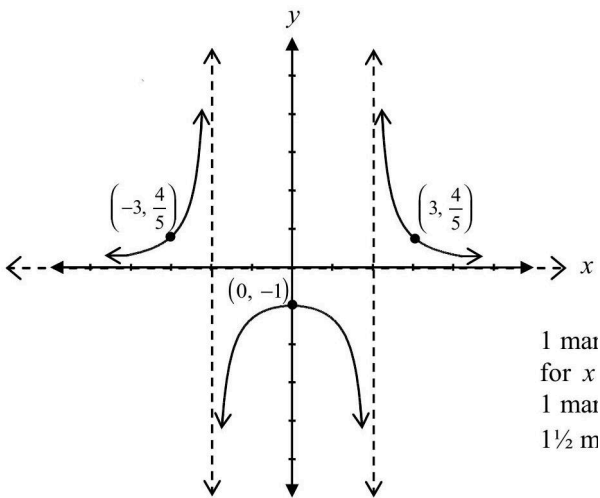
Sketch the graph of the function $f(x)$ and determine the y -intercept.

$$f(x) = \frac{4}{(x - 2)(x + 2)}$$



y -intercept:

Solution



-1

$\frac{1}{2}$ mark for y -intercept

4 marks

Question 27

Kim solved the following logarithmic equation:

$$\log_2\left(-\frac{x}{3}\right) = \log_2(x - 4)$$

$$-\frac{x}{3} = x - 4$$

$$-x = 3x - 12$$

$$-4x = -12$$

$$x \leq 3$$

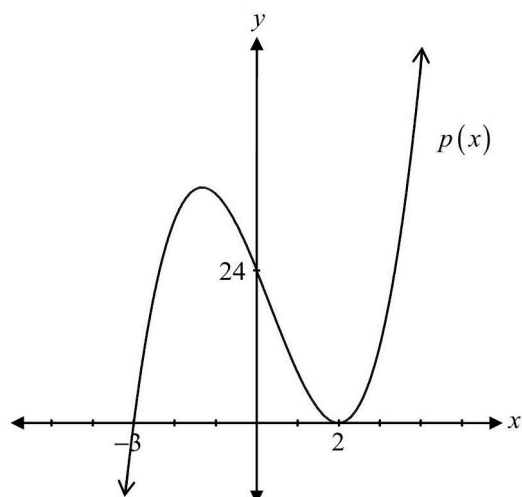
Explain why $x = 3$ is an extraneous solution.

Solution

$x = 3$ is an extraneous solution because the argument in a logarithmic equation cannot be negative.

Question 28

Determine the equation of the polynomial function represented by the graph.



$p(x) =$

Solution

$$p(x) = a(x - 2)^2(x + 3)$$

$$24 = a(0 - 2)^2(0 + 3)$$

$$24 = a(4)(3)$$

$$24 = 12a$$

$$a = 2$$

$$p(x) = 2(x - 2)^2(x + 3)$$

1 mark for x -intercepts ($\frac{1}{2}$ mark for each)

1 mark for multiplicity at $x = 2$

1 mark for leading coefficient

3 marks

Question 29

Determine the coterminal angles with $\frac{2\pi}{3}$ over the interval $[-2\pi, 4\pi]$.

Solution

$$-\frac{4\pi}{3}, \frac{8\pi}{3} \quad 1 \text{ mark (} 1/2 \text{ mark for each coterminal angle)}$$

1 mark

Question 30

a) Solve the following equation:

$$0 = \sqrt{4x - 8} - 2$$

b) Explain how your answer in a) is related to the graph of $y = \sqrt{4x - 8} - 2$.

Solution

a)

$$4 = 4x - 8$$

$$12 = 4x$$

$$x = 3, x \geq 2$$

b) The answer in a) is the x -intercept of the graph.

- $x \geq 2$ does not need to be shown.

Question 31

Determine the exact value of $\sin \frac{13\pi}{12}$.

Solution

$$\begin{aligned}\sin \frac{13\pi}{12} &= \sin \left(\frac{3\pi}{4} + \frac{\pi}{3} \right) \\ &= \left(\sin \frac{3\pi}{4} \right) \left(\cos \frac{\pi}{3} \right) + \left(\cos \frac{3\pi}{4} \right) \left(\sin \frac{\pi}{3} \right) \\ &= \left(\frac{\sqrt{2}}{2} \right) \left(\frac{1}{2} \right) + \left(-\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

1 mark for combination

2 marks for exact values ($\frac{1}{2}$ mark for each)

3 marks

- Other combinations are possible.

Question 32

a) 2 marks

b) 1 mark

Given the functions $f(x) = x + 2$ and $g(x) = \frac{1}{x-5}$:

a) Determine the equation of the composite function $f(g(x))$ and its domain.

$$f(g(x)) =$$

domain:

b) Determine the x -intercept and y -intercept of $f(g(x))$.

x -intercept:

y -intercept:

Solution

a) $f(g(x)) = \frac{1}{x-5} + 2$

or

$$f(g(x)) = \frac{2x-9}{x-5}$$

domain: $\{x \in \mathbb{R}, x \neq 5\}$

b) x -intercept:

$$\frac{9}{2}$$

$\frac{1}{2}$ mark for x -intercept

y -intercept:

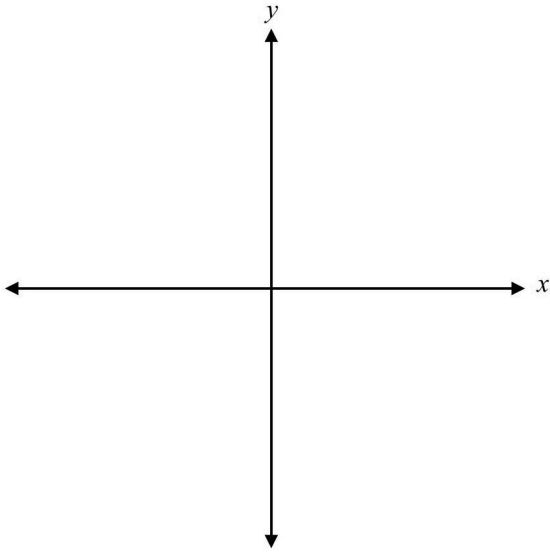
$$\frac{9}{5}$$

$\frac{1}{2}$ mark for y -intercept

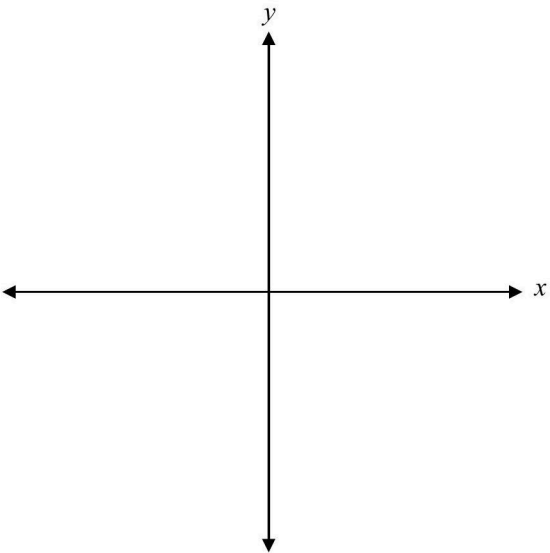
1 mark

Question 33

a) Sketch the graph of $f(x) = \log_5(x - 1)$.



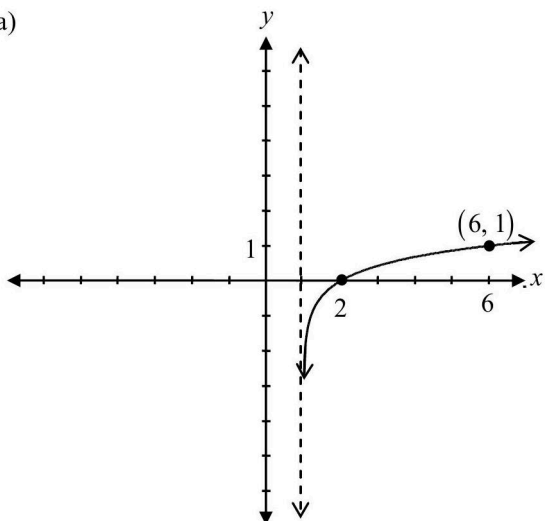
b) Sketch the graph of $f^{-1}(x)$.

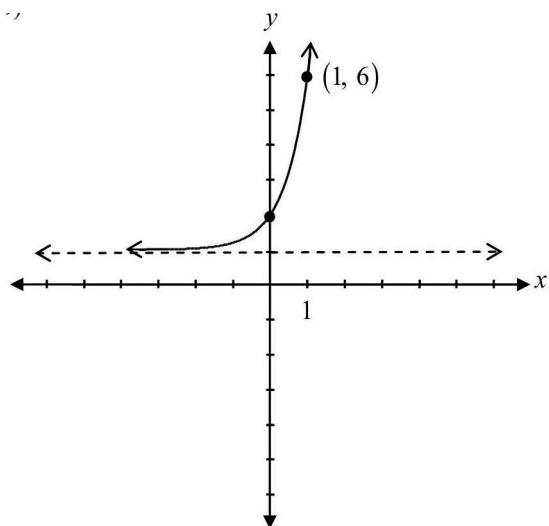


Solution

b)

a)





$\frac{1}{2}$ mark for vertical asymptote at $x = 1$

$\frac{1}{2}$ mark for x -intercept at $x = 2$

$\frac{1}{2}$ mark for increasing logarithmic function

$\frac{1}{2}$ mark for consistent point on the

logarithmic graph

2 marks

1 mark for the graph of the inverse

function consistent with a)

1 mark

Question 34

Explain what the graph of a rational function looks like near a vertical asymptote.

Solution

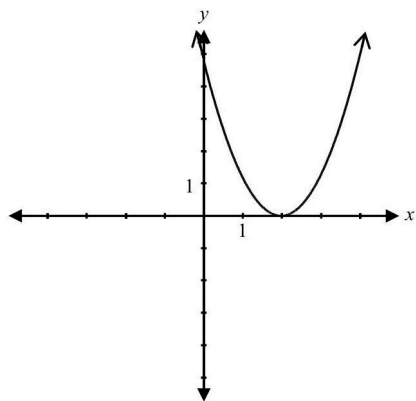
The graph approaches infinity (positive or negative) as it approaches the asymptote.

or

The graph approaches the asymptote, but does not touch it.

Question 35

Given the graph of $f(x) = (x - 2)^2$,



determine one possible restriction for the domain of $f(x)$ so that its inverse is a function.

Domain:

Solution

$[2, \infty)$

$[2, \infty)$

Domain:

$(-\infty, 2)$

- Other solutions are possible.

Question 36

Over the interval $[0, 2\pi]$, determine the non-permissible values of θ in the expression

$$\csc \theta (\cos \theta + 1).$$

Solution

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sin \theta \neq 0$$

$$\theta = 0, \pi, 2\pi$$

1 mark for correct substitution of an appropriate identity

$\frac{1}{2}$ mark for $\sin \theta \neq 0$

$\frac{1}{2}$ mark for consistent non-permissible values

2 marks

Question 37

Explain why ${}_3C_8$ is undefined.

Solution

In the formula ${}_nC_r$, the number of objects, n , must be larger than or equal to the number of objects selected, r .

or

You can't select 8 objects from a total of 3.

Question 38

Solve:

${}_nP_3 = 48(n - 1)$

Solution

$\frac{n!}{(n-3)!} = 48(n - 1)$
$\frac{(n)(n-1)(n-2)(n-3)!}{(n-3)!} = 48(n - 1)$
$n(n - 2) = 48$
$n^2 - 2n - 48 = 0$
$(n - 8)(n + 6) = 0$
$n = 8 \quad n > -6$

$\frac{1}{2}$ mark for substitution in correct formula

1 mark for expansion of factorial

$\frac{1}{2}$ mark for simplification of factorials

$\frac{1}{2}$ mark for solving for both values of n

$\frac{1}{2}$ mark for rejecting extraneous solution

3 marks

Question 39

b) 1 mark

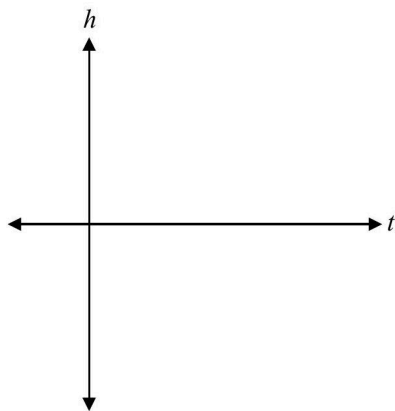
c) 1 mark

Christine dives off a diving board.

Her dive is modelled by the function $h(t) = t^3 - 3t^2 - t + 3$, where h is her height in metres, relative to the water surface and t is the time in seconds after diving off the diving board.

a) Given that $(t + 1)$ is a factor for the function $h(t)$, determine the other factors.

b) Sketch the graph of the function $h(t)$ for the time interval $t = 0$ to $t = 3$.



c) Determine how many seconds Christine is underwater.

Solution

a)

-1	1	-3	-1	3
↓		-1	4	-3
1		-4	3	0
$h(t)$		$(t + 1)$	$) (t^2$	$-4t + 3)$
$h(t)$		$(t + 1)$	$)(t$	$1) (t - 3)$

b)

c) Christine is underwater for 2 seconds.

$\frac{1}{2}$ mark for $t = -1$

1 mark for synthetic division (or equivalent strategy)

$\frac{1}{2}$ mark for determining other factors

2 marks

$\frac{1}{2}$ mark for t -intercepts

$\frac{1}{2}$ mark for h -intercept

1 mark

1 mark

Question 40

Prove the identity for all permissible values of x .

$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

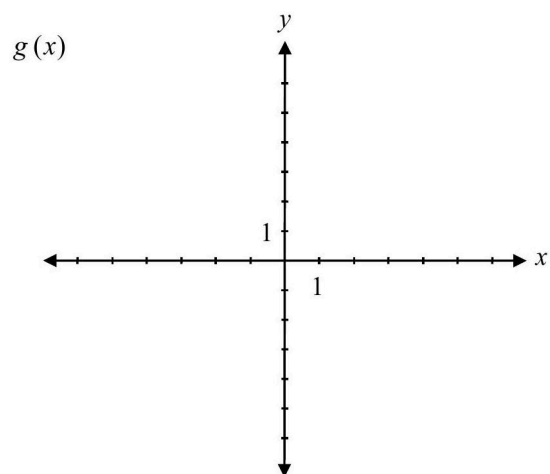
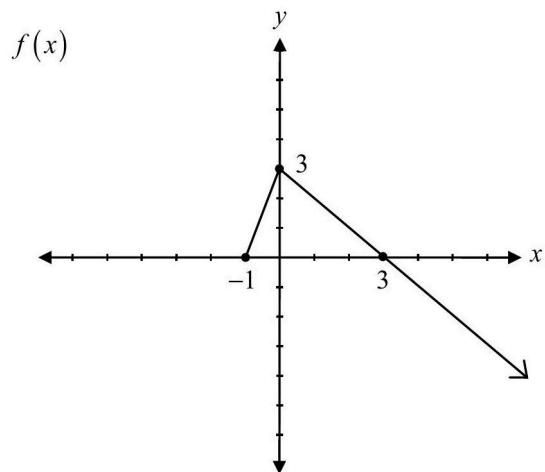
Left-Hand Side	Right-Hand Side

Solution

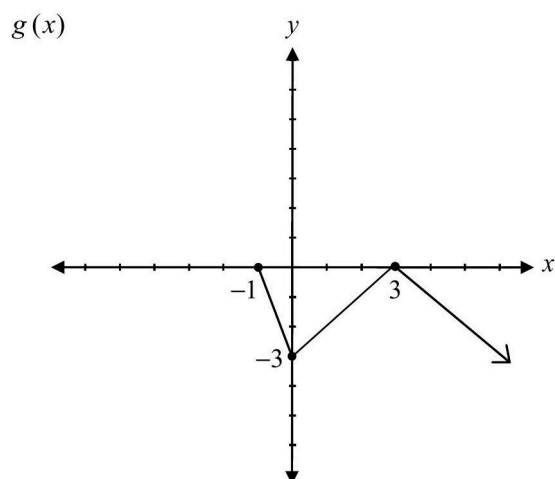
Method 1		
	Left-Hand Side	Right-Hand Side
	$\frac{\sec x + \tan x}{\cos x} + \frac{\sin x}{\cos x}$ $\frac{(1 + \sin x)(1 - \sin x)}{\cos x(1 - \sin x)}$ $\frac{1 - \sin^2 x}{(1 - \sin x) \cos x}$ $\frac{\cos^2 x}{(1 - \sin x) \cos x}$ $\frac{\cos x}{1 - \sin x}$	1 mark for correct substitution of appropriate identities 1 mark for logical process to prove the identity 1 mark for algebraic strategies 3 marks

Question 41

Given the graph of $f(x)$, sketch the graph of the function $g(x) = -|f(x)|$.



Solution



1 mark for absolute value

1 mark for vertical reflection

2 marks

Question 42

Given that $\cot \theta = -\frac{2}{5}$, where θ is in Quadrant IV, determine the exact value of $\sin \theta$.

Solution

$$\begin{array}{llll} P(2, -5) & & & \\ r^2 & = (2)^2 + (-5)^2 & & \\ r^2 & = 4 + 25 & \frac{1}{2} \text{ mark for substitution of } x = 2, y = -5 & \\ r & = \sqrt{29} & 1 \text{ mark for } \sin \theta \left(\frac{1}{2} \text{ mark for quadrant, } 1/2 \text{ mark for value } \right) & \end{array}$$

2 marks